## Exercise sheet: Week 3

1. Let $\underline{X}$ be a random sample of size $n$ from the exponential distribution with p.d.f.

$$f_X(x;\theta) = \frac{1}{\theta}\exp\left(-\frac{x}{\theta}\right)$$

for $x > 0$. Show that the score function can be written as

$$U(\theta;\underline{X}) = \frac{n}{\theta^2}(\bar{X} - \theta).$$

Deduce that $\bar{X}$ is the MLE of $\theta$. What else can you deduce about the estimator $\bar{X}$ from the theory that we studied this week (see MVUE Lemma)?

2. Let $\underline{X}$ be a random sample of size $n$ from the exponential distribution with p.d.f.

$$f_X(x;\lambda) = \lambda\exp\left(-\lambda x\right)$$

for $x > 0$. This is equivalent to the previous question, with the parameterisation $\lambda = \frac{1}{\theta}$.

   (a) Show that the score function can be written as

$$U(\lambda;\underline{X}) = n\left(\frac{1}{\lambda} - \bar{X}\right).$$

   Explain why there is no unbiased estimator that attains the CRLB.

   (b) Show that the MLE for $\lambda$ is

$$\hat{\lambda} = \frac{1}{\bar{X}},$$

   find its expectation and find the value of $c$ such that $c\hat{\lambda}$ is an unbiased estimator of $\lambda$.

   (c) Find the variances of $\hat{\lambda}$ and also $c\hat{\lambda}$, and compare with the CRLB for unbiased estimators of $\lambda$, in particular considering how they behave as $n \to \infty$.

Hints: Before starting part (b) and (c), show that $\bar{X} \sim$ Gamma$(n, n\lambda)$, and that for $Y \sim$ Gamma$(\alpha, \beta)$, $E(Y^{-k}) = \frac{\Gamma(\alpha-k)}{\Gamma(\alpha)}\beta^k$ and then use these properties in your solution.

3. Let $\underline{X}$ be a random sample from a distribution with p.d.f.

$$f_X(x;\sigma^2) = \frac{x}{\sigma^2}\exp\left(-\frac{x^2}{2\sigma^2}\right), x > 0.$$

Consider the cases (a) $\theta = \sigma^2$ and (b) $\theta = \sigma$.

In each case, determine the score function $U(\theta;\underline{X})$ and deduce whether or not the CRLB is attainable. In both cases determine the CRLB.